

Notes: Mayer, Morrison, Piskorski, and Gupta (AER, 2014)

Mortgage Modification and Strategic Behavior: Evidence from a Legal Settlement with Countrywide

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 The Countrywide settlement
- 2.2 Treatment and control groups
- 2.3 Data sources
- 2.4 Sample restrictions
- 2.5 Main outcome: the rollover rate
- 2.6 Important borrower and loan variables

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Core difference-in-differences idea
- 3.2 Main probit specification
- 3.3 Event-study version
- 3.4 Why the sample ends in February 2009
- 3.5 Heterogeneity by liquidity and equity
- 3.6 Nontargeted-debt tests
- 3.7 Back-of-the-envelope policy calculation

4 Identification and instruments (what makes the estimates credible)

- 4.1 The main identification assumption
- 4.2 Why the settlement is useful for identification
- 4.3 Comparability of treatment and control loans

4.4	Threat 1: Countrywide loans may be intrinsically riskier
4.5	Threat 2: the effect may reflect general economic distress
4.6	Threat 3: borrowers may simply accelerate inevitable defaults
4.7	Threat 4: HAMP and other modification programs
4.8	Threat 5: borrower moral or financial deterioration across all debts
4.9	No formal instruments
5	Tables and figures
5.1	Figure 1: Internet searches for “Countrywide Modification”
5.2	Table 1: Summary statistics for 2/28 ARMs
5.3	Figures 2 and 3: Kernel densities of observables
5.4	Figure 4: Evolution of default rates for 2/28 ARMs
5.5	Table 2: Baseline default specification for 2/28 ARMs
5.6	Table 3: Heterogeneity by credit utilization and CLTV
5.7	Table 4: Nontargeted debts
5.8	Additional robustness checks
6	Short summary
7	Conclusion

1 Big picture

- **Question.** Do homeowners strategically default when a mortgage modification program makes delinquency a key eligibility requirement?
- **Setting.** The paper studies the October 2008 legal settlement between Countrywide Financial Corporation and US state attorneys general. Countrywide agreed to offer loan modifications to certain Countrywide-serviced subprime first-lien mortgage borrowers who were seriously delinquent or near delinquent.
- **Why the setting is useful.** The settlement was publicly announced before the main program rollout, applied nationally to Countrywide-serviced loans, and made delinquency central to eligibility. These features create a sharp information shock: some borrowers suddenly learned that becoming delinquent could help them qualify for mortgage relief.
- **Economic trade-off.** Simple eligibility rules can deliver help quickly and cheaply, but they are manipulable. More rigorous screening may reduce strategic defaults, but it is slower and may fail to prevent avoidable foreclosures. The paper estimates one cost of simple rules: induced default.
- **Main design.** The authors compare Countrywide-serviced subprime 2/28 adjustable-rate mortgages with similar non-Countrywide mortgages before and after the settlement announcement. The main outcome is the monthly rate at which borrowers go straight from current to 60 days delinquent.
- **Main result.** Countrywide's monthly rollover rate rises by about **0.48–0.54 percentage points** after the announcement, roughly a **10–11 percent** increase relative to the pre-settlement rollover rate.
- **Strategic-behavior evidence.** The effect is strongest among borrowers who looked least likely to default soon: borrowers with substantial unused credit card liquidity and borrowers with current CLTV below 100. These patterns are hard to explain purely by worsening distress.
- **Placebo logic.** The settlement targeted subprime first mortgages. The paper does not find comparable increases in defaults on credit cards, second liens, or non-subprime fixed-rate mortgages. This supports the interpretation that the rise is tied to the modification incentive rather than a general Countrywide borrower shock.
- **Takeaway.** Mortgage modification programs with simple delinquency-based eligibility rules can create meaningful strategic behavior. But this does not automatically imply that such programs are bad: the policy question is whether the cost of induced default is smaller or larger than the benefits from faster relief and fewer foreclosures.

2 Data and measurement (key objects)

2.1 The Countrywide settlement

In June 2008, California and Illinois sued Countrywide over alleged deceptive lending practices. Many other state attorneys general followed. On October 6, 2008, Countrywide reached a multistate settlement.

The settlement targeted **subprime first mortgages serviced by Countrywide**, especially hybrid adjustable-rate mortgages and option ARMs. A loan could qualify if:

- it was originated before 2008 and was in Countrywide’s servicing portfolio on June 30, 2008;
- the borrower’s loan-to-value ratio was at least 75 percent;
- the borrower was at least 60 days delinquent, or was likely to become delinquent because of an interest-rate reset or negative amortization trigger;
- and post-modification mortgage payments satisfied program thresholds.

For seriously delinquent subprime hybrid ARMs, the program could restore the introductory interest rate for five years and could offer fully amortizing modifications with reduced rates, in some cases as low as 3.5 percent.

Important timing. The settlement was announced in October 2008, while Countrywide planned to begin contacting eligible borrowers in December 2008. This gap is central: borrowers had time to respond to the news before program implementation.

2.2 Treatment and control groups

The treatment group is composed of mortgages serviced by **Countrywide**. The control group consists of comparable mortgages serviced by other servicers that were not subject to the Countrywide settlement.

The main analysis focuses on **2/28 ARMs**. These are subprime-style adjustable-rate mortgages with an initial two-year period followed by reset. They are natural to study because they were central to the settlement and to the subprime crisis.

The paper also studies **non-subprime fixed-rate mortgages** as a placebo-like group. These mortgages were much less likely to be targeted by the settlement, so a broad Countrywide-specific distress shock should appear there too, while a settlement-induced incentive effect should not.

2.3 Data sources

The paper matches three main data components:

- **BlackBox Logic:** loan-level mortgage data for privately securitized subprime, Alt-A, and prime loans. These data include origination characteristics, servicer name, interest rate, term, balance, payment status, and monthly delinquency status.
- **Equifax:** borrower-level monthly credit report data. These data include Vantage credit score, mortgage and installment debt, revolving debt balances, credit limits, and credit card utilization.
- **Zillow zip-level house price indices:** used to update property values and construct current combined loan-to-value ratios.

The BlackBox-Equifax match is performed by 1010Data. The authors further restrict the matched sample by requiring the Equifax borrower zip code to match the property zip code in BlackBox, improving match quality and focusing on owner-occupants.

2.4 Sample restrictions

The base sample is restricted to loans that could plausibly be relevant to the settlement:

- first-lien residential mortgages on owner-occupied primary residences;
- loans originated in 2005, 2006, or the first half of 2007;
- mortgages with origination LTV at least 70;
- and loans not serviced by Citibank, IndyMac, or J.P. Morgan, because those institutions had contemporaneous modification programs.

After these restrictions, the base sample contains more than **500,000 2/28 ARMs** and more than **700,000 fixed-rate mortgages**. The matched sample contains more than **300,000 2/28 ARMs** and more than **450,000 fixed-rate mortgages**.

2.5 Main outcome: the rollover rate

The key dependent variable is whether a borrower becomes 60 days delinquent in month t , conditional on being current two months earlier:

$$Y_{it} = 1\{\text{loan } i \text{ is 60+ days delinquent in month } t \mid \text{current in } t - 2\}. \quad (1)$$

The paper calls this the **rollover rate**: the rate at which borrowers “roll straight” from current to 60 days past due.

This outcome is well suited to the question because:

- the settlement made 60-day delinquency a key eligibility threshold;
- abrupt default from current status is more suggestive of a response to new incentives;
- and conditioning on previously current loans compares relatively similar borrowers.

2.6 Important borrower and loan variables

The controls include origination and current loan characteristics, borrower credit variables, and geography. The key variables are:

- origination FICO and current Vantage credit score;
- origination LTV and current LTV;
- origination CLTV and current CLTV;
- current and initial mortgage interest rates;
- loan balance;
- origination quarter;
- mortgage reset timing;

- MSA fixed effects;
- credit card utilization and available credit;
- and junior-lien balances.

A central heterogeneity variable is **available credit card liquidity**. The paper often measures this as the number of monthly mortgage payments the borrower could cover with unused credit card lines. Another central variable is current **CLTV**, which proxies for how far above or below water the borrower is.

3 Models and empirical specifications (all equations + interpretation)

3.1 Core difference-in-differences idea

The empirical question is whether Countrywide borrowers became more likely to default after the settlement announcement relative to similar non-Countrywide borrowers.

The cleanest conceptual difference-in-differences estimand is:

$$\delta = \left[\text{Default}_{\text{CW, post}} - \text{Default}_{\text{CW, pre}} \right] - \left[\text{Default}_{\text{Control, post}} - \text{Default}_{\text{Control, pre}} \right]. \quad (2)$$

Interpretation. This removes common default-rate shocks that hit all similar mortgages and isolates the differential post-announcement change among Countrywide-serviced loans.

3.2 Main probit specification

The paper estimates a probit model of the form:

$$\mathbb{P}(Y_{it} = 1 \mid \text{Current}_{i,t-2} = 1) = \Phi \left(\text{CW}_{it}\beta + \text{OctDec}_{it}\mu + \text{CW}_{it} \times \text{OctDec}_{it}\delta + x'_{it}\gamma \right). \quad (3)$$

Here:

- Y_{it} equals one if loan i is 60 days delinquent in month t , conditional on being current two months earlier;
- CW_{it} equals one for loans serviced by Countrywide;
- OctDec_{it} equals one in October–December 2008;
- $\text{CW}_{it} \times \text{OctDec}_{it}$ is the treatment interaction;
- x_{it} includes borrower, loan, time, reset, and geographic controls;
- and $\Phi(\cdot)$ is the standard normal CDF.

The coefficient of interest is δ . The paper reports marginal effects from the probit, so a coefficient such as 0.0054 means a **0.54 percentage point** increase in the monthly rollover probability.

3.3 Event-study version

The actual implementation is richer than the simple one-post-period equation. The authors include dummies for each quarter around the settlement and interact them with the Countrywide indicator:

$$\mathbb{P}(Y_{it} = 1 \mid Current_{i,t-2} = 1) = \Phi \left(\alpha + CW_i \beta + \sum_q \lambda_q Q_{qt} + \sum_q \delta_q (CW_i \times Q_{qt}) + x'_{it} \gamma \right). \quad (4)$$

The omitted period is July–September 2007. The key post-period is October–December 2008. The authors also report tests comparing the settlement quarter directly to the quarter immediately before it, July–September 2008.

Interpretation. The pre-settlement interactions test whether Countrywide loans were already trending differently. The post-settlement interaction tests whether Countrywide rollover rates jump exactly when the settlement could change incentives.

3.4 Why the sample ends in February 2009

The baseline analysis runs from July 2007 to February 2009. July 2007 roughly marks the collapse of the subprime refinancing market. February 2009 is chosen to avoid contamination from the federal Home Affordable Modification Program, which was announced in February and implemented in March 2009.

Interpretation. The authors want the post-settlement effect to reflect the Countrywide settlement, not later national modification policy.

3.5 Heterogeneity by liquidity and equity

A pure distress story predicts larger effects among borrowers who are already close to default. A strategic-behavior story can predict a strong response among borrowers who are not otherwise likely to default but can gain by becoming eligible for relief.

The paper therefore estimates the same specification separately for:

- borrowers with unused credit card liquidity greater than five months of mortgage payments;
- borrowers with unused liquidity between one and five months;
- borrowers with unused liquidity of zero to one month;
- borrowers with current CLTV below 100;
- borrowers with current CLTV between 100 and 120;
- borrowers with current CLTV above 120;
- and underwater borrowers with high available credit.

Interpretation. If the treatment effect is largest among liquid or above-water borrowers, the evidence is more consistent with strategic response than with a simple worsening-balance-sheet explanation.

3.6 Nontargeted-debt tests

The settlement targeted subprime first-lien mortgages. The authors therefore estimate similar models for:

- second-lien mortgage delinquency;
- credit card delinquency;
- and non-subprime fixed-rate mortgage delinquency.

Interpretation. If the Countrywide settlement merely coincided with a broad shock to Countrywide borrowers, delinquencies should rise on many debts. If it changed incentives tied to first-lien mortgage modification eligibility, the increase should appear mainly on targeted first mortgages.

3.7 Back-of-the-envelope policy calculation

The authors use the estimated effect to illustrate potential economic magnitude. The baseline estimate is a 0.54 percentage point monthly effect for three months:

$$3 \times 0.0054 = 0.0162. \quad (5)$$

Applied to roughly 45 million current first-lien mortgages, this implies:

$$45,000,000 \times 0.0162 \approx 729,000 \quad (6)$$

additional defaults under a national program with similar simple eligibility rules.

If a typical modification forgives value equal to 30 percent of a 200,000 balance, the lender-side cost per unnecessary modification is about:

$$0.30 \times 200,000 = 60,000, \quad (7)$$

which produces a total cost of about:

$$729,000 \times 60,000 \approx 43.7 \text{ billion dollars.} \quad (8)$$

Interpretation. Even a modest monthly effect can matter in aggregate because the mortgage market is enormous.

4 Identification and instruments (what makes the estimates credible)

4.1 The main identification assumption

The key assumption is that, absent the Countrywide settlement, comparable Countrywide and non-Countrywide loans would have followed similar rollover-rate trends after October 2008.

This is a parallel-trends assumption adapted to a setting with loan-level controls and a discrete policy announcement.

4.2 Why the settlement is useful for identification

The settlement helps identification because it has several quasi-experimental features:

- **Unexpected public announcement.** Borrowers learned about a new modification opportunity around October 6, 2008.
- **Eligibility linked to delinquency.** Being at least 60 days past due was central to qualification.
- **Treatment tied to servicer, not borrower choice.** Borrowers did not choose Countrywide servicing in response to the settlement.
- **Control group exists.** Similar mortgages serviced by other firms were not covered by the Countrywide settlement.
- **Program timing.** Announcement came before rollout, giving a window in which borrowers could adjust behavior.

4.3 Comparability of treatment and control loans

The paper shows that Countrywide and control 2/28 ARMs are similar in average observables before the settlement. In September 2008, for example:

- initial CLTV is 92 for Countrywide versus 90 for control loans;
- current CLTV is 121 versus 119;
- initial interest rate is 8.18 percent versus 8.07 percent;
- current interest rate is 8.63 percent versus 8.56 percent;
- origination FICO is 618 versus 617;
- current Vantage score is 667 versus 666;
- and credit card utilization is 0.47 in both groups.

There is, however, somewhat more mass in the high-risk tail of the Countrywide distribution by September 2008. The empirical strategy deals with this by adding detailed controls and by checking that pre-settlement differential default patterns disappear after accounting for high-risk loans.

4.4 Threat 1: Countrywide loans may be intrinsically riskier

A concern is that Countrywide loans might have become riskier for reasons unrelated to the settlement. The evidence addresses this in several ways:

- adding detailed BlackBox and Equifax controls reduces pre-settlement Countrywide differences;
- trimming to a more homogeneous sample removes visible pre-settlement differential default patterns;

- the main effect is concentrated in October–December 2008, right after the settlement announcement;
- and the effect is stronger among low-risk-looking borrowers, not only among high-risk borrowers.

4.5 Threat 2: the effect may reflect general economic distress

The authors examine borrowers with substantial unused credit card liquidity and borrowers with CLTV below 100. These borrowers were less likely to default in the near term absent the settlement. Yet their rollover rates respond strongly.

Interpretation. A general distress shock should be strongest among borrowers with little liquidity or deeply underwater mortgages. The opposite heterogeneity pattern is important evidence for strategic behavior.

4.6 Threat 3: borrowers may simply accelerate inevitable defaults

The authors extend the analysis through December 2009. If the settlement merely pulled forward defaults that would have occurred soon anyway, later Countrywide default rates should fall below control rates. The paper does not find negative and significant later interactions.

Interpretation. The settlement appears to increase the stock of delinquent Countrywide loans, not just shift the timing of inevitable defaults by a few months.

4.7 Threat 4: HAMP and other modification programs

HAMP could confound later estimates because it affected borrowers broadly and may have interacted differently with Countrywide loans. The baseline stops before HAMP implementation. The authors also exclude servicers with contemporaneous modification programs from the control group.

4.8 Threat 5: borrower moral or financial deterioration across all debts

If the settlement caused a broad collapse in borrower repayment discipline, or if Countrywide borrowers simply experienced worse shocks, defaults should also rise on credit cards and second liens. The paper finds no such increase.

In some cases, credit card delinquency even declines for Countrywide borrowers with high available credit. This is consistent with borrowers stopping first-mortgage payments while using the extra cash flow to remain current on other debts.

4.9 No formal instruments

The paper does not rely on an instrumental variable. Its credibility comes from the settlement-induced treatment timing, difference-in-differences comparisons, detailed loan-level controls, pre-trend diagnostics, and placebo outcomes.

5 Tables and figures

5.1 Figure 1: Internet searches for “Countrywide Modification”

Read: Internet search volume for “Countrywide Modification” spikes sharply around October 6, 2008, the day the settlement was announced.

Economic message: The settlement was salient. Borrowers had a plausible channel through which to learn that Countrywide modification benefits were available and that delinquency mattered for eligibility.

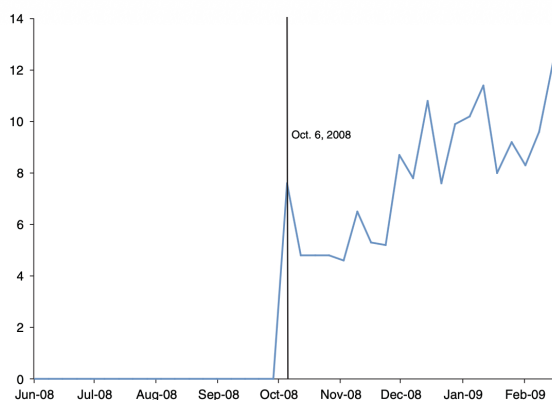


FIGURE 1. INTERNET SEARCHES FOR “COUNTRYWIDE MODIFICATION”

Notes: This figure plots an index of the weekly volume of Internet searches for the term “Countrywide Modification,” as reported by Google Trends. Searches for this term spiked on October 6, 2008 the day the Countrywide settlement was announced and reported by newspapers around the country.

TABLE 1—SUMMARY STATISTICS FOR 2/28 ARMS

	Countrywide		Control group	
	43,418		203,960	
Number of loans	Mean	SD	Mean	SD
<i>Panel A. BlackBox variables</i>				
Initial CLTV	92	8.8	90	8.9
Initial interest rate	8.18	1.28	8.07	1.24
Current interest rate	8.63	1.35	8.56	1.37
Origination FICO	618	54	617	56.3
Origination balance	196	114	206	133
Low/no doc	0.42	0.49	0.6	0.49
Refi	0.055	0.23	0.1	0.3
Cash out refi	0.39	0.49	0.37	0.48
<i>Panel B. Equifax variables</i>				
Current CLTV	121	28	119	28
Current Vantage	667	82	666	84
2nd lien balance	47.9	31.4	52.4	38.4
Has junior lien	0.44	0.5	0.38	0.48
Credit utilization	0.47	0.37	0.47	0.37

Notes: This table presents summary statistics for 2/28 adjustable rate mortgages (ARMs) serviced by Countrywide and the control group in the matched sample as of September 2008. The summary statistics include characteristics of these loans at origination and as of September 2008, the month before public announcement of the settlement on October 6, 2008. CLTV and interest rates are reported in percentage terms; loan balances are in thousands of dollars.

5.2 Table 1: Summary statistics for 2/28 ARMs

Read: In the matched sample as of September 2008, Countrywide and control loans look similar on average. Countrywide has 43,418 loans and the control group has 203,960. Initial CLTV is 92 versus 90; current CLTV is 121 versus 119; origination FICO is 618 versus 617; current Vantage is 667 versus 666; and credit utilization is 0.47 in both groups.

Economic message: The treatment and control groups are not obviously incomparable at the mean. This makes the difference-in-differences design more credible, though the distributional figures show that risky tail differences still need attention.

5.3 Figures 2 and 3: Kernel densities of observables

Read: Figure 2 shows that in July 2007, Countrywide and control distributions of interest rates, Vantage scores, and CLTV look very similar. Figure 3 shows that by September 2008, the distributions remain broadly similar, but Countrywide has somewhat more mass in high-risk tails, especially high CLTV and high interest rates.

Economic message: The main identification concern is not mean imbalance but tail risk. This motivates the detailed controls and the robustness checks that trim extreme-risk observations.

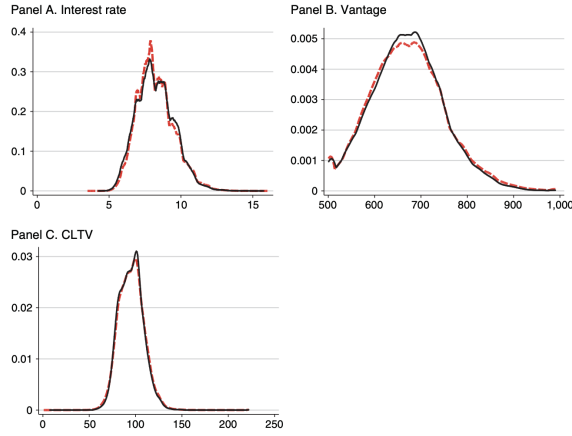


FIGURE 2. COMPARABILITY OF COUNTRYWIDE AND CONTROL GROUP: KERNEL DENSITY OF OBSERVABLES, JULY 2007

Notes: This figure shows the kernel density plots as of July 2007 for mortgage interest rate (panel A), Vantage credit score (panel B), and CLTV (panel C) for 2/28 ARMs in the matched sample. The solid line shows the Countrywide loans while the dashed line shows the corresponding values for mortgages in the control group.

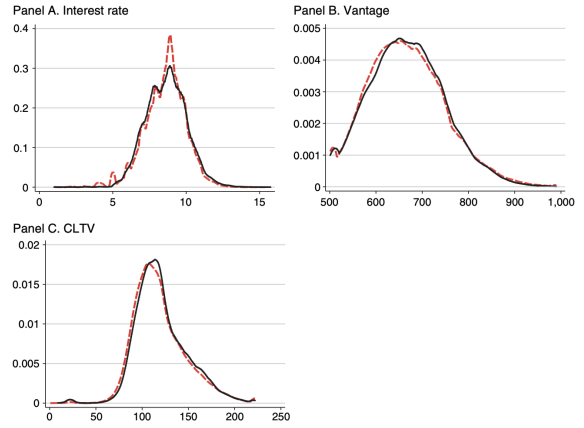


FIGURE 3. COMPARABILITY OF COUNTRYWIDE AND CONTROL GROUP: KERNEL DENSITY OF OBSERVABLES, SEPTEMBER 2008

Notes: This figure shows the kernel density plots as of September 2008 for mortgage interest rate (panel A), Vantage credit score (panel B), and CLTV (panel C) for 2/28 ARMs in the matched sample. The solid line shows the Countrywide loans while the dashed line shows the corresponding values for mortgages in the control group.

5.4 Figure 4: Evolution of default rates for 2/28 ARMs

Read: The Countrywide rollover rate jumps relative to the control group in October–December 2008. For all loans, there is also some pre-settlement divergence in July–September 2008. But the pre-period issue is weaker for borrowers with high available credit and largely absent for borrowers with CLTV below 100.

Economic message: The timing of the jump lines up with the settlement announcement. The cleaner patterns among liquid and above-water borrowers strengthen the strategic-default interpretation.

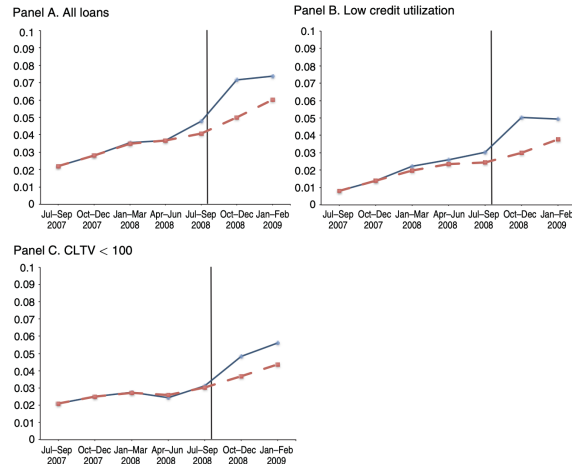


FIGURE 4. EVOLUTION OF DEFAULT RATES FOR 2/28 ARMs: COUNTRYWIDE AND CONTROL GROUP

Notes: This figure plots the average monthly rollover rate (the 60-day delinquency rate among borrowers who were current two months before) during each of the five quarters preceding the settlement announcement, the quarter just after the announcement (Oct–Dec 2008), and the Jan–Feb 2009 period. Panel A plots the rate for all the loans, panel B subsets on borrowers with low credit utilization, and panel C subsets on borrowers whose mortgages had a CLTV less than 100 percent at the time of delinquency. The solid line shows the rollover rate for Countrywide loans while the dashed line shows the corresponding rate for mortgages in the control group.

TABLE 2—DEFAULT SPECIFICATION FOR 2/28 ARMS

	Base sample (1)	Base sample (2)	Matched sample (3)	Matched sample (4)
Countrywide × Oct–Dec 2007	−0.0012** (0.0005)	−0.0005 (0.0005)	0.0002 (0.0006)	0.0004 (0.0005)
Countrywide × Jan–Mar 2008	−0.0003 (0.0005)	−0.0008 (0.0005)	−0.0002 (0.0006)	0.0002 (0.0005)
Countrywide × Apr–Jun 2008	−0.0004 (0.0005)	−0.0022*** (0.0005)	−0.0018*** (0.0006)	−0.0014** (0.0006)
Countrywide × Jul–Sep 2008	0.0056*** (0.0006)	−0.0003 (0.0005)	0.0001 (0.0006)	0.0006 (0.0006)
Countrywide × Oct–Dec 2008	0.0160*** (0.0007)	0.0052*** (0.0006)	0.0048*** (0.0007)	0.0054*** (0.0006)
Countrywide × Jan–Feb 2009	0.0092*** (0.0008)	0.0001 (0.0007)	−0.0005 (0.0007)	0.0005 (0.0007)
Countrywide	−0.0002 (0.0004)	0.0016*** (0.0003)	0.0017*** (0.0004)	0.0018*** (0.0004)
Origination quarter	No	Yes	Yes	Yes
BlackBox control	No	Yes	Yes	Yes
MSA control	No	Yes	Yes	Yes
Reset control	No	Yes	Yes	Yes
Equifax control	No	No	No	Yes
Number of cases	9,448,457	9,448,457	6,261,055	6,261,055
Avg. delinquency 2008:III	0.049	0.049	0.048	0.048
Average share Countrywide	0.15	0.15	0.18	0.18
Countrywide × (2008:IV–2008:III)	0.0104	0.0055	0.0047	0.0048
Wald Test (p-value)	0.0000	0.0000	0.0000	0.0000

Notes: This table reports estimates of probit specification (1) using data on Hybrid 2/28 ARMs. The dependent variable takes the value of one when a loan becomes 60 days past due in a given month, conditional upon being current 60 days earlier, and is equal to zero otherwise. Column 1 estimates the model using the base sample and includes only time dummies, a Countrywide dummy that equals one if the loan is serviced by Countrywide, and interactions between the Countrywide and time dummies. The excluded category is July–September 2007, the first quarter of our analysis period. Column 2 adds indicators for the quarter of origination and additional controls from the BlackBox data, including a wide range of loan and borrower-level characteristics, such as origination LTV, origination FICO and CLTV (when available), and their interactions with time dummies, initial interest rate, current interest rate, reset controls capturing the timing of reset, and MSA fixed effects for the location of the property backing the loan. Columns 3 and 4 use the matched sample instead of the base sample. Column 3 includes the same controls as in column 2; column 4 includes additional Equifax controls, such as current CLTV, credit card utilization, and current credit scores (Vantage) and their change over time. Coefficients reported are marginal effects from a probit regression; standard errors are in parentheses; standard errors are clustered at the loan ID.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

5.5 Table 2: Baseline default specification for 2/28 ARMs

Read: Across specifications, the Countrywide $\times$ October–December 2008 interaction is positive, statistically significant, and economically meaningful. With controls, the effect is about **0.48–0.54 percentage points**. The pre-settlement rollover rate is about 4.8 percent.

Economic message: The settlement increases the monthly probability of rolling straight from current to 60 days delinquent by about **10–11 percent**. Adding controls and moving from the base sample to the matched sample does not eliminate the effect.

Important diagnostic: In the minimal-control model, there is a pre-settlement July–September 2008 Countrywide increase. Once borrower, loan, reset, MSA, and Equifax controls are added, this pre-settlement increase becomes small and statistically insignificant.

5.6 Table 3: Heterogeneity by credit utilization and CLTV

Read: The settlement effect is present across liquidity and equity groups, but the relative effect is largest among borrowers who seemed less likely to default otherwise. Borrowers with more than five months of available credit show a 0.54 percentage point increase, about an 18 percent rise relative to their pre-settlement rate. Borrowers with CLTV below 100 show a 0.50 percentage point increase, about a 16 percent rise. Underwater borrowers with high available credit show a 0.49 percentage point increase, about a 12 percent rise.

Economic message: The effect is not driven only by the most distressed borrowers. The strongest relative responses appear among borrowers with liquidity or positive equity, which is exactly where strategic response is most plausible.

TABLE 3—DEFAULT SPECIFICATION FOR 2/28 ARMS BY CURRENT CREDIT UTILIZATION AND CLTV

	> 5 months (1)	1–5 months (2)	0–1 months (3)	CLTV < 100 (4)	100 ≤ CLTV < 120 (5)	120 ≤ CLTV > 100 (6)	> 5 months CLTV > 100 (7)
Countrywide $\times$ Oct–Dec 2007	−0.0004 (0.0008)	0.0008 (0.0009)	−0.0001 (0.0005)	0.0011 (0.0006)	−0.0011 (0.0008)	−0.0059 (0.0035)	−0.0026 (0.0016)
Countrywide $\times$ Jan–Mar 2008	0.0012 (0.0009)	0.0008 (0.0009)	−0.0001 (0.0005)	0.0013 (0.0006)	−0.0008 (0.0008)	−0.0054 (0.0034)	−0.0006 (0.0016)
Countrywide $\times$ Apr–Jun 2008	0.0003 (0.0007)	−0.0014 (0.0009)	−0.0014 (0.0010)	−0.0009 (0.0007)	−0.0032*** (0.0008)	−0.0055 (0.0033)	−0.0024 (0.0015)
Countrywide $\times$ Jul–Sep 2008	0.0011 (0.0008)	0.0018 (0.0010)	0.0015 (0.0011)	0.0009 (0.0008)	−0.0011 (0.0009)	0.0007 (0.0031)	−0.0011 (0.0015)
Countrywide $\times$ Oct–Dec 2008	0.0054*** (0.0011)	0.0052*** (0.0012)	0.0063*** (0.0013)	0.0050*** (0.0011)	0.0038*** (0.0012)	0.0076 (0.0039)	0.0049*** (0.0021)
Countrywide $\times$ Jan–Feb 2009	0.0002 (0.0009)	0.0009 (0.0012)	0.0015 (0.0014)	0.0049*** (0.0011)	−0.0004 (0.0011)	−0.0029 (0.0035)	−0.0028 (0.0017)
Countrywide	−0.0001 (0.0005)	0.0011 (0.0007)	0.0025*** (0.0008)	0.0005 (0.0004)	0.0040*** (0.0007)	0.0078** (0.0035)	0.0025 (0.0016)
Origination quarter	Yes	Yes	Yes	Yes	Yes	Yes	Yes
BlackBox control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MSA control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reset control	Yes	Yes	Yes	Yes	Yes	Yes	No
Equifax control	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,210,922	1,639,789	1,748,890	1,994,158	1,715,891	1,067,641	611,797
Avg. share Countrywide	0.18	0.19	0.18	0.16	0.18	0.19	0.20
Avg. Countrywide 2008:III delinquency	0.030	0.041	0.052	0.031	0.039	0.078	0.040
Countrywide $\times$ (2008:IV–2008:III)	0.0043	0.0034	0.0048	0.0041	0.0049	0.0069	0.006
Wald Test (p-value)	0.0000	0.0019	0.0001	0.0010	0.0000	0.0001	0.0000

Notes: This table reports estimates of probit specification (1) for Hybrid 2/28 ARMs, but stratifies the Matched Sample by borrowers current credit utilization and CLTV. The models are estimated using the full set of BlackBox and Equifax controls used in column 4 of Table 2. The dependent variable takes the value of one if the loan becomes 60 days past due in a given month, conditional upon being current 60 days earlier, and is equal to zero otherwise. The excluded category is July–September 2007, the first quarter of our analysis period. Columns 1 through 3 separate borrowers by available credit card utilization, 4 through 6 separate them by CLTV, and 7 subsets on underwater borrowers with high available credit utilization. Coefficients reported are marginal effects from a probit regression; standard errors are in parentheses; standard errors are clustered at the loan ID.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

TABLE 4—DEFAULT SPECIFICATIONS FOR OTHER DEBT TYPES

	5+ months utilization				CLTV < 100			Non-subprime FRM (7)
	2nd lien (1)	Credit card (2)	2nd lien (3)	Credit card (4)	2nd lien (5)	Credit card (6)		
Countrywide $\times$ Oct–Dec 2007	0.0002 (0.0008)	−0.0016 (0.0012)	−0.0011 (0.0012)	−0.0026 (0.0017)	0.0001 (0.0006)	−0.0009 (0.0020)	0.0000 (0.0000)	
Countrywide $\times$ Jan–Mar 2008	0.0012 (0.0009)	0.0009 (0.0014)	−0.0009 (0.0011)	−0.0030 (0.0018)	0.0011 (0.0011)	0.0039 (0.0025)	0.0003*** (0.0000)	
Countrywide $\times$ Apr–Jun 2008	0.0000 (0.0000)	0.0004 (0.0015)	−0.0018 (0.0010)	−0.0022 (0.0020)	−0.0002 (0.0010)	0.0013 (0.0030)	0.0003*** (0.0001)	
Countrywide $\times$ Jul–Sep 2008	0.0004 (0.0008)	−0.0001 (0.0007)	−0.0016 (0.0011)	−0.0016 (0.0020)	0.0013 (0.0014)	−0.0000 (0.0000)	0.0000 (0.0000)	
Countrywide $\times$ Oct–Dec 2008	0.0004 (0.0009)	−0.0018 (0.0016)	−0.0018 (0.0011)	−0.0047** (0.0021)	−0.0002 (0.0012)	0.0011 (0.0037)	−0.0002*** (0.0000)	
Countrywide $\times$ Jan–Feb 2009	−0.0003 (0.0081)	−0.0016 (0.0019)	−0.0031*** (0.0010)	−0.0031 (0.0026)	0.0017 (0.0019)	0.0071 (0.0051)	−0.0002*** (0.0000)	
Countrywide	0.0018** (0.0008)	−0.0014 (0.0012)	0.0033** (0.0014)	0.0003 (0.0017)	0.0001 (0.0005)	−0.0026 (0.0018)	0.0004*** (0.0000)	
Origination quarter	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
BlackBox control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MSA control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Reset control	Yes	Yes	Yes	Yes	Yes	Yes	No	
Equifax control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,906,151	3,230,307	525,191	903,902	265,800	870,999	8,920,874	0.39
Avg. share Countrywide	0.20	0.18	0.21	0.18	0.18	0.16	0.16	
Avg. Countrywide 2008:III delinquency	0.028	0.073	0.021	0.045	0.010	0.082	0.090	
Countrywide $\times$ (2008:IV–2008:III)	0.0000	−0.0017	−0.0002	−0.0031	−0.0015	0.0011	−0.0002	
Wald Test (p-value)	0.94	0.28	0.83	0.12	0.31	0.77	0.0000	

Notes: This table reports estimates from models similar to probit specification (1), but with the dependent variable measuring delinquency on loans that were not targeted by the settlement. The models are estimated using the matched sample and the full set of BlackBox and Equifax controls used in column 4 of Table 2. In column 1 below, the dependent variable equals one if the borrower becomes 60 days past due on a second lien, conditional upon being current two months earlier, and equals zero otherwise. In column 2, the dependent variable equals one if the borrower becomes delinquent on credit card debt, conditional upon being current two months earlier, and equals zero otherwise. Columns 3–6 show the corresponding results after subsetting on borrowers with low credit utilization (columns 3 and 4) and CLTV < 100 (columns 5 and 6). Column 7 uses data on borrowers with fixed-rate mortgages (FRMs). The dependent variable equals one if the borrower becomes 60 days past due on a FRM, conditional upon being current two months earlier, and equals zero otherwise. Across all columns, the excluded category is July–September 2007, the first quarter of our analysis period. Coefficients reported are marginal effects from a probit regression; standard errors are in parentheses; standard errors are clustered at the loan ID.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

5.7 Table 4: Nontargeted debts

Read: There is no positive post-settlement jump in defaults on second liens, credit cards, or non-subprime fixed-rate mortgages. The key October–December 2008 estimates are close to zero or

negative: for example, second-lien delinquency is about 0.0004, credit card delinquency is about -0.0018, and non-subprime FRM delinquency is about -0.0002.

Economic message: The settlement effect is specific to targeted first mortgages. Borrowers do not appear to default broadly across all debts; they adjust behavior on the debt category that determines modification eligibility.

5.8 Additional robustness checks

Extended analysis period: The authors extend the sample through December 2009 and do not find later negative Countrywide effects that would offset the initial jump. This weakens the “mere acceleration” explanation.

Alternative hazard model: A discrete-time hazard model produces similar results, showing that the rollover specification is not mechanically driving the findings.

Alternative control groups: Adding interactions for the largest non-Countrywide servicers and using Wells Fargo as a placebo servicer do not reproduce the Countrywide pattern.

Monthly time effects: Replacing quarterly time effects with monthly interactions still shows a meaningful increase in November and December 2008.

6 Short summary

Mayer, Morrison, Piskorski, and Gupta study whether borrowers strategically default when mortgage modification eligibility depends on delinquency. Their setting is the 2008 Countrywide legal settlement, which promised modification offers to seriously delinquent Countrywide-serviced subprime mortgage borrowers.

The empirical design compares Countrywide 2/28 ARM borrowers with similar non-Countrywide borrowers before and after the October 2008 settlement announcement. The outcome is a sharp default transition: going from current to 60 days delinquent within two months.

The main estimate is that Countrywide borrowers become about 0.5 percentage points more likely per month to roll straight into 60-day delinquency after the announcement. Relative to a pre-settlement default rate around 4.8 percent, this is about a 10 percent increase.

The strongest evidence for strategic behavior comes from heterogeneity and placebo tests. Borrowers with substantial unused credit and borrowers with CLTV below 100 show especially large relative responses, while defaults do not rise on credit cards, second liens, or non-subprime fixed-rate mortgages.

The policy implication is a trade-off. Simple delinquency-based modification programs may create strategic defaults, but more careful screening can be slow and may allow preventable foreclosures. The paper quantifies one side of this trade-off rather than claiming that all generous modification programs are undesirable.

7 Conclusion

Core conclusion. The Countrywide settlement increased mortgage delinquency among covered borrowers. The increase is concentrated exactly where the program changed incentives and is especially strong among borrowers who were not obviously forced into default by immediate distress.

Economic intuition. When eligibility for relief is tied to delinquency, delinquency becomes an option value. Some borrowers who could continue paying may stop paying to access modification benefits. This is the household-finance version of moral hazard in debt relief.

Identification lesson. The paper is persuasive because it combines a sharp policy announcement, a servicer-based treatment group, comparable control loans, rich borrower-level controls, pre-trend checks, heterogeneity by financial slack, and placebo outcomes on debts that were not targeted.

Policy lesson. Faster, simpler mortgage modification rules can prevent costly foreclosures, but they can also induce unnecessary default. More rigorous screening can reduce strategic behavior, but it delays relief and may miss borrowers who need help quickly.

Takeaway. The paper does not say that mortgage modification is bad. It says that program design matters: eligibility rules shape borrower behavior, and policymakers must weigh the cost of strategic default against the benefits of rapid, broad debt relief.